

Collaborative Design of Augmented Flashcards for Design History Class

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ABSTRACT

This case study looks at augmented reality (AR) flashcards used in a university-level classroom. The collaborative design effort used RtD and pulled from the outcomes of three prior case studies using AR in the classroom. This poster focuses on the latest case study. Using research through design (RtD) to develop various AR user interface (UI) for educators to use as a framework to augment classroom content aiming to improve learning outcomes for all students. The case study shows that more research is warranted, and there is a viable need for using AR in higher education humanities courses.

Keywords: Augmented Reality. Education. Design. User experience.

Index Terms:

1 INTRODUCTION

Augmented reality can be used in higher education to engage students and provide access to content in unique ways. AR has also been shown to increase motivation [2,3]. Augmented reality has the unique ability to blend the physical and digital. Students can have access to physical artifacts that act as portals to digital content layers. Augmented reality also offers pedagogical scenarios that provide many affordances [4, 5, 6, 7, 8], including digital learning supports. A series of case studies conducted after completing Augmented reality reading support in higher education: Exploring effects on perceived motivation and confidence in comprehension for struggling readers in higher education, look at the use of augmented content in university classrooms. This paper focuses on the latest case study exploring the use of augmented flashcards in a university-level design history classroom. This case study used augmented flashcards that were designed based on three previous case studies and one usability study where augmented content was used in higher education design classes.

Using AR in education is not a new topic and there are several literature reviews looking at AR in education [4, 5, 9, 10]. Much literature on AR in education focuses on k-12 not higher education and AR in education has been used for STEM courses with less research conducted with humanities classes. Experiments with AR in art museums are common, but AR in art education is not as prevalent. This case study focuses on using AR at the university level for humanities classes like Design History. The Case study shows a desire for students to have access to AR in the classroom. Issues of how the AR is accessed and content creation, especially 3D models, remain hurdles to mass implementation.

2 THEORY

Augmented Flash Cards can provide an immersive digital learning experience through familiar study artifacts to improve engagement with study materials. AR creates hybrid immersive learning environments that facilitate critical thinking, problem-solving, and communication [14]. Technology can aid students and improve their digital literacy skills. This research is significant to all educators and students. New research needs to occur to understand augmented reality benefits for classrooms as AR technology changes.

2.1 Case Study Background

This Case Study of two graphic design history class will use mixed methods and RtD to explore using augmented flashcards with added digital resources. The study looked at the perceived motivation and support the AR flashcards provided. Students also provided qualitative feedback on their attitude toward using AR in the classroom and their perception of AR influence on the learning outcomes to see if additional digital tools added through augmented reality can improve learning outcomes for all students. While all students stand to benefit, struggling readers may benefit the most from alternative learning aids. For this study, no measurement of the students reading ability will be taken prior to the study.

2.2 Method

For this case study, augmented flashcards were created through a group design activity in GD159 Immersive Design at Fresno State University. The class reviewed Usability challenges for augmented flashcards and study guides brought up in previous studies as well as observations from their own frustrations with augmented content. A design style guide was created for the printed version of the flashcards and template for the Augmented Content was also created to keep the cards consistent. Despite the use of a group style guide, some variation in design and UI, especially on the augmented digital layer, were present. This included variation of available resources for each flashcard topic.



Figure 1: Example of AR flashcards using Zappar App codes.

The study was conducted in two separate GD 135 Design History classes in Fall 2019 at the authors institution. The classes had 18 and 19 students enrolled. All 37 students were able to participate and use the augmented flash cards 21 chose to take the optional survey and 17 finished the survey.

Students were given printed flashcards the class before exam 3; the same time they would normally receive the normal paper study guide. After receiving the flash cards the students needed to fold each card in half to make the correct front and back set up for the flashcards. They were given time in class to review the flashcards and access the augmented reality through their smartphones. No in-depth directions or tutorials were given on how to use the AR. Students were able to take the flashcards home and use them as study aids before the 3rd exam. After the exam was complete, students were given a link to an optional survey to provide feedback on the

experience. They also took the reduced IMMS to look at the motivation of the AR flashcards.

2.3 Results

The results of the survey showed that students overall thought the augmented reality was helpful and would or would consider using it again if it was available. The major complaints were inconsistencies in the linked digital resources the augmentation provided, the size of phone screens used to access the AR and preference for writing out their own flashcards to study. When asked if they would use AR cards in the future 10 said yes, 6 maybe and 1 no. When asked if the AR was simple to navigate 16 said yes, and 2 no. When asked if they would be interested in learning new concepts using AR cards 9 said yes, 7 maybe, and 1 no.

3 DISCUSSION

Augmented reality (AR) allows content to be experienced through multiple mediums that can motivate and engage students. This article defines AR as a Medium wherein digital information overlays the physical world, dependent on the perspective of the individual interacting with and experiencing the AR Medium [12, 1, 13, 3].

The use of virtual reality (VR) and augmented reality (AR) in the classroom over the last few decades show the potential to enhance learning through the combination of virtual and physical worlds [8]. Using AR in the classroom is becoming increasingly common thanks to the processing power of mobile devices. Smartphones offer the ability to access augmented digital content, use AR as a portal to more immersive web based virtual reality experiences or even social virtual reality experiences that can allow for collaboration or group study sessions. This case study showed there is indeed interest in using augmented reality flashcards. Improvements needed include the use of 3D models or 360 photos where appropriate. Creating a framework that allows for a scaffold immersion from a traditional physical artifact of a flash card to an augmented viewing of additional digital information all the way to a completely immersive virtual environment will allow students to choose the level of XR they want to access. Allowing for access to social virtual environment can also aid students in collaborating or holding group study sessions. These improvements along with considerations from the following limitations section will allow for a new study that improves the framework of AR flashcards and adapts them for remote learning.

3.1 Limitations

While this study was conducted in two separate design history classes only 21 students started the post activity survey. Of the 21 surveys started only 17 were completed. The flashcards were provided in the class before the test when the study guide is normally handed out. Due to the in-depth nature of the flashcards providing the students access earlier may have increased the participation rate. When asked what would make the AR flashcards more interesting or helpful one student stated, "receiving them a week before the test." Another limitation was the format of the flashcard the students received. The cards were printed on Strips of paper that needed to be folded in half to complete the two-sided flashcard. Some students found this tedious and did not want to fold their own cards resulting in a lower participation rate.

4 CONCLUSION

Overall students are interested in the use of AR in a classroom; however, there are existing technology issues that hold it back. The use of handheld devices makes the technology accessible but cumbersome. Faculty can use simple AR creation tools to augment existing content but this process can be time-consuming. In order for AR to benefit the most people it needs to be simple for faculty to

generate the augmented content and less cumbersome for the user to view. The use of smart glasses could potentially greatly assist. Another important component is using digital content to its full advantage and making sure all digital content is ADA accessible.

Future research should look into the use of AR for Humanities based courses in higher education. Continuing to usability test different ways of providing digital content with physical content will also be important. Especially as technology allows for wearable smart glasses or contacts instead of handheld devices. Finally, studies looking at the learning outcomes and retention of using AR-enhanced learning compared to traditional learning methods will be important.

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